

M002 Extension 5 — Redstone AND Solver

Topic: AND — two redstones both true · **Course:** Maze Madness · **Level:** Extension (Beginner) · **Time:** about 30 minutes

Same **Week 3 world**. Find the **hardest maze** your teacher shows a picture of — some tiles need **two** redstones true at once (an **AND**), so type `run` to start.

1 · Chat Words

A chat word makes the agent move.

```
type "l":  
  turn left
```

Type `l` . **Circle one:** left · right · jump

```
type "r1":  
  come back to you
```

Type `r1` . **Circle one:** to goal · back to you

2 · Both Must Be True

Rule #1 needs **two** redstones. It runs only when **both** are true.

```
redstone on left AND redstone below:  
  move up 1, turn right
```

Redstone on left. Nothing below. **Does it run? Circle one:** yes · no

Redstone on left AND redstone below. **Does it run? Circle one:** yes · no

3 · Spot the Trick 🐉

Rule #1 needs **both** redstones. This one checks only the left.

```
# clean – needs BOTH  
redstone on left AND redstone below:  
    move up 1, turn right
```

```
# buggy – only checks left  
redstone on left:  
    move up 1, turn right
```

The buggy one runs even when there is **no** redstone below. **Why is that wrong? Circle one:** it needs both redstones · it needs no redstone · it turns twice

4 · Fill the Gap

Run the rule only when **both** redstones are there. One word is missing.

```
redstone on left ____ redstone below:  
  move up 1, turn right
```

Missing word? Circle one: AND · IF · OR

5 · Show Your Work 📷 🎥

Open the Week 3 world and find the maze from the picture. Type **run** and watch the agent reach the **diamond** goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.